Learning Objective 6.2: I can decode DTMF tones from a captured audio signal.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). The keypad grid: rows 697, 770, 852, and 941 Hz carry 1 2 3 A, 4 5 6 B, 7 8 9 C, and * 0 # D; columns are 1209, 1336, 1477, and 1633 Hz, left to right.

1. **Reading the grid.**

   (a) Which two tones does the '8' key send? The '#' key?

   8:

   #:

   (b) A receiver's bins show 941 Hz and 1209 Hz in one digit and 770 Hz and 1633 Hz in the next. Which two keys were pressed?

   (c) In one block, the 697 Hz and 770 Hz bins are both strong and about equal, and so is the 1336 Hz bin. Which test of the L13 decoder rejects the block, and what could cause a signal like that?

2. **Timing.** An auto-dialer sends each digit as 70 ms of tone followed by 30 ms of silence.

   (a) How many digits a second does it send, and how long does an 11-digit number take, from the first tone to the end of the last one?

   rate: ____

   time: ____

   (b) Does this dialer meet the standard's timing rules from Lesson 13? If not, which rule does it break, and what can go wrong at a receiver built only to the standard?

3. **A shorter Goertzel block.** A designer tries Goertzel blocks of $N = 136$ samples at $f_s = 8$ kHz instead of the lesson's 205.

   (a) Find the block time and the spacing of the bins.

   block: ____

   spacing: ____

   (b) Find k for 852 Hz. Round it to an integer and find the frequency of that bin and its error from 852 Hz, in Hz and in percent.

   k: ____

   bin: ____

   error: ____

   (c) Explain why that rounding is a problem for a DTMF receiver, and what the L13 decoder does instead.

   (d) With bins 58.8 Hz apart, is the 73 Hz spacing of 697 and 770 Hz past the first null of each bin's response? What does that mean for the 6 dB test?

4. **Twist.** A capture of the '9' key has a row tone of amplitude 0.20 and a column tone of amplitude 0.12.

   (a) Find the twist in dB. Does the standard require a receiver to accept it?

   (b) In `L13-dtmf-rx.grc`, each stock Goertzel block feeds a Complex to Mag, which shows half a tone's amplitude. What heights do the 852 Hz and 1477 Hz traces reach?

   852:

   1477:

   (c) Telephone lines lose more at high frequencies, so many phones send the column tone about 2 dB above the row tone. If the line loses 6 dB more at the column tone than at the row tone, what twist arrives? Name the case the standard calls this.

5. **The decision rules.** The L13 decoder scales each bin's power so that a lone tone gives 1, and the eight values below are one block's shares of its energy.

Bin (Hz)	697	770	852	941	1209	1336	1477	1633
Block A	0.02	0.41	0.03	0.01	0.01	0.02	0.38	0.02
Block B	0.12	0.10	0.02	0.01	0.03	0.30	0.02	0.01

   (a) Apply the three tests to block A: the two-bin share, each winner's margin over the rest of its group, and the twist. Which digit does block A hold?

   (b) Which test does block B fail?

   (c) Block A alone does not produce a digit. What else must happen before the decoder reports '6', and what does that requirement reject?

6. **Noise.** The L13 decoder reads every number correctly down to an SNR of about 2 dB, measured across the whole 0 to 4 kHz band.

   (a) With $N = 205$, the bins are 39.0 Hz apart. About how much less noise does one bin see than the whole 4 kHz band? Give the answer in dB.

   (b) At 0 dB SNR the bins still see the tones about 20 dB above their noise, yet the decoder fails. Explain in words which test stops it.